My passion: Open source and motorsports
My definition of open source: Sharing and spreading of knowledge
Myth about open source: It is software that is FREE
One thing I like about open source: The mailing list
India and open source: Some major projects have been contributed from India, today

Harris, a New Zealand based developer, had developed Pegasus mail for some anonymous corporation. We were using it for free with reasonably good support, though the source code was in his hand. We could recommend changes as per our needs and, if approved, we had to pay a small amount commensurate with the additional effort needed to make those changes. Later, after we rolled out Linux networking, we shifted to Sendmail and hosted our own internal mailing servers.

The irony is that, today, we want the best software but we hardly use 10 per cent of it. Big bosses always want the best machines but since most of them are not technically proficient, they under-utilise them. To address this issue, I devoted quality time in conducting a series of workshops with a few corporates, on the optimal usage of even basic software like word processing, teaching them about mail merge to improve efficiency. I aggressively participated in events that promoted the effective use of office applications.

Open source is not just about software that is FREE

This is a popular myth and I laugh when people's only definition of open source is that it is free. The fact is that it is usually easier to obtain than proprietary software, often resulting in its increased use. Sometimes open source helps developers to build loyalty and experience a sense of ownership of the end product. I can say that the open source software (OSS) development approach offers the potential for more flexible technology and quicker innovation.

Open source is all about code, so even if the original developer stops working on it, the code continues to exist and be developed by its users. Also, it uses open standards that are accessible to everyone; thus, it does not come with the problem of incompatible formats that exist in the proprietary software world.

In a nutshell, in open source, 'free' does not mean zero cost, but freedom—the freedom to use, freedom to change and the freedom to redistribute.

> *"The mailing list and its later avatar, the forum, are all about sharing knowledge. It is like a broadcast that gets you possible solutions to specific problems."*

Better results in spite of budget constraints

Just to share an example, at one point we required 120 desktops in our company. Let's, calculate the typical cost of such infrastructure—the hardware would have cost ₹ 60,000-70,000 per machine. And then we required Windows, for email and an office suite, which meant another ₹ 30,000 – 35,000 per system for just software. It all added up to a huge budget. Since we had decided not to use any unlicensed software, open source or Linux was the only viable solution to cut down costs.

But what about the hardware? Fortunately, we came across an open source project called LTSP (Linux Terminal Server Project). The founder and project leader of LTSP, Jim McQuillan, was targeting computer learning in schools at the lowest cost. This model uses a free and open source terminal server for Linux that allows many people to simultaneously use the same powerful server. Applications run on the server with a terminal known as a thin client (also known as an X terminal) handling input and output. Generally, terminals are low-powered, lack a hard disk, and are quieter and more reliable than desktop computers because they do not have any moving parts.

Using this concept, we could easily run 20-25 terminals off one computer. Instead of buying one computer for everyone, we could use older generation computers as terminals and only invest in two or three new computers. This meant big savings of around 60 per cent on hardware too.

This technology later gained popularity in schools as it allowed them to provide students access to computers without purchasing or upgrading expensive desktop machines. Improving access to computers becomes less costly as thin client machines can be older computers that are no longer suitable for running a full desktop OS. Even a relatively slow CPU with as little as 128MB of RAM can deliver excellent performance as a thin client. In addition, the use of centralised computing resources means that more performance can be gained for less money through upgrades to a single server rather than across a fleet of computers. We started with M4Linux at the time of K12 LTSP. It was a startup to promote the adoption of open source. In fact, here we implemented a few thin client architecture networks for clients.

In 1990, we had started a plant at a new location about 250 kilometres from our base in Delhi. At that time, we were sold the concept of a paperless office by a well known IT firm, that charged huge licence costs and a high maintenance fee, and sold us costly hardware. Yet, neither was the firm able to deliver speed nor handle real-world volumes of data. The solution also couldn't cope with changes in parameters like excise duty structures, new cesses, etc, which meant going back to the developers for changes. And changes always meant extra bills to be paid. Finally, in desperation,